NOT MEASUREMENT SENSITIVE
MIL-P-85856/31
7 December 1988

MILITARY SPECIFICATION SHEET

POWDER MATERIALS FOR FLAME AND PLASMA SPRAYED COATINGS TUNGSTEN CARBIDE-COBALT, TYPE II, CLASS 2, COMPOSITION A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the powder described herein shall consist of this document and the latest issue of MIL-P-85856.

Chemical Composition

Fe	C	Co	W
2.0 max.	3.6 4.3	10.0 13.0	80.0 min.

(1) Determination not required for routine acceptance.

Particle Size/Distribution

+200 Mesh (+75 µm)	5.0 max.
+325 Mesh (+45 µm)	90.0 min.
-325 Mesh (-45 µm)	10.0 max.

Metallographic Inspection

Tungsten carbide-cobalt particles shall be angular in shape in the as-received condition and shall be predominantly 20 microns or less in size when examined at a magnification of 250 to 500x.

Sprayed Coating Properties

Bond Strength	45.0 MPa (6500 psi) min.
Hardness	Rockwell 15N, 85 min.
Microstructure	See paragraph 4.5.4.1.2 Porosity - 3% max. Unmelted Particles - 10% max.

AMSC N/A FSC 3439
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Deposit Efficiency and Deposition Rate

30% min for A/H_2 @ 4.0 lb/hr

Comparable Documents

Powder which meets the requirements specified is qualified under the following specifications.

Pratt & Whitney Aircraft PWA 1302

Custodians:
Air Force - 20
Navy - AS
Army - MR

Review activities:
Air Force - 99, 71
Navy - SH

Preparing activity:
Navy - AS
(Project No. 3439-0663/31)

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